From System Ranking to Decision Value: How Much Human Evaluation Do Automatic Evaluators Actually Save?

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Abstract

Automatic evaluators are credited with the human labels they save, usually by correlation with human scores. We measure the credit for the decision an evaluation is run for: certifying from a human audit that a chosen system is within ε of the best of a few close competitors. It counts unique human labels with and without the evaluator, after a judge-free annotation design (deduplication, known-zero differences skipped) is credited first. On WMT22 MQM, LMArena and retrieval the design saves a median 4.6% of uniform-sampling labels over 30 close MT decisions; the mean metric adds 2.3% refitted and nothing frozen, the strongest metric of each year 4–5%. The reason is the level of measurement: metrics with system-level correlation 0.93–0.99 correlate only 0.11–0.38 with the paired differences a close decision depends on; the saving is about $\rho^2(1 - B_{\text{pilot}}/J)$. Strategies fixed on WMT22 and applied to WMT23 give the rule: deduplicate, add one strong evaluator fixed in advance, refit its coefficient (3% cheaper than the design under the unrestricted protocol, 0.4% once certificates wait for budgets where the bound's error is nominal); choosing the evaluator from a pilot is no better (-0.1%, $[-0.7, +0.5]$), and freezing the coefficient costs its value.

1 Introduction

Human judgments remain the reference for comparing NLP systems, and they are expensive. Automatic evaluators promise to stretch them: score everything automatically and spend human effort only on correcting the evaluator's bias, as in prediction-powered inference (PPI; Angelopoulos et al., 2023a,b). The saving is now measured directly (Boyeau et al., 2025; Chen et al., 2026), and Gao et al. (2026) rank MT metrics by the annotation they save (PPSR): under uniform labelling, with the coefficient fitted on the evaluation labels, the strongest WMT22 metrics save about 9–15% of the labels a paired comparison needs.

We do not revisit that number. We ask what is left of it for the decision an evaluation is run for, and how an auditor should use it. A team has a short list of strong systems that are close in quality and wants to certify from a human audit that the one it picks is within ε of the best. Three things separate this from the asymptotic picture. The decision is *local*: the systems that matter are the top few, not the field over which system-level correlation is computed. The audit need not sample *uniformly*: items whose paired difference is known to be zero (identical outputs) need no label, and such labels are saved without any evaluator, which should be credited only with what it adds on top. And the coefficient that corrects the evaluator must be *estimated* from the audit's own labels, either frozen after a small pilot or refitted as labels arrive.

Our method is correspondingly plain. We simulate the audit on data whose human labels exist, count the unique human labels it needs to certify the decision, and define an evaluator's human-evidence saving (HES) as the share of those labels it removes on top of the judge-free design (§2). A concrete case fixes the bookkeeping. Two translation systems are to be compared on 1,000 source segments. On the segments where both produce the same string the quality difference is zero and nothing need be rated; on the rest, a pilot of 50 segments is rated for both systems, the candidate is the pilot leader, and a pre-committed number of further segments is rated; the decision is certified if a one-sided bound on the competitor's lead is then below ε. We report the budget at which half of the simulated audits certify, not a sequential rule that stops when the bound first crosses ε. An evaluator enters as a control variate: its score differences predict the human ones, so each rated segment is worth more. Its value is the labels it removes from this audit, *after* the identical segments have been skipped and *including* the pilot, which it cannot touch. The pilot matters: if the audit needs 250 labels and the pilot

is 100 of them, an evaluator that cuts the remaining 150 by 10% saves 6% of the total (hypothetical numbers). Three findings follow (§3).

1. **The design comes first; the average evaluator adds little, the strongest as much again.** Over 30 two-system WMT22 decisions (2,000 simulated audits per cell, 31 metric variants) the judge-free design saves a median 4.6% of uniform-sampling labels (8–16% in the four pre-registered cells, 41–56% in retrieval); on the same design the mean metric adds 2.3% with a refitted coefficient and 0.2% with a frozen one, and a one-sided bound rules out a 10% increment for 94% of decision–evaluator pairs. The strongest metric of each year (MetricX-XXL on WMT22, XCOMET-XXL on WMT23) adds 4–5%, more than the design itself in a third to over half of the cells.

2. **The reason is the level of measurement.** Metrics with system-level Pearson 0.93–0.99 correlate only 0.11–0.38 with the paired differences a close decision depends on, and the realised saving is about $\rho^2(1 - B_{\text{pilot}}/J)$: the squared decision-level correlation, diluted by the pilot's share of the cost (§4).

3. **A pre-fixed strong evaluator is as good as choosing one from the pilot.** Four strategies fixed on WMT22 and applied unchanged to WMT23 (42 metrics) give the same answer in both years: dedup plus one strong evaluator fixed in advance, with a refitted coefficient, costs 3% less than the design alone (in all 26 informative WMT23 cells); choosing the evaluator from the pilot instead changes the cost by -0.1% $[-0.7, +0.5]$ in a paired comparison at 2,000 audits per cell, and costs 4% more when the coefficient is frozen. Under boundary stress the bound's type-I error is above nominal below this experiment's minimum budgets (about 206 and 265 labels on the two language pairs), for the human-only design as much as for the evaluators; a protocol that certifies only beyond them keeps 0.4% of the evaluator's saving on the same cells (3.0% at a tolerated error of 0.125), because most of the 30 decisions certify earlier (§3.3).

PPSR answers which evaluator saves most; we ask what the saving is worth once the decision is local, the design is credited first and the coefficient is fitted on the audit. We propose no estimator: the corrected estimator is PPI++ with a control variate, the allocation is active inference (Zrnic and Candès, 2024a) for the decision's functional. Three pre-registered locks fixed the hypotheses before evaluator outputs were joined with human labels; three of ten failed and are reported (App. F).

2 The decision and its cost

Decision. A menu $\mathcal{M}$ of systems is evaluated on a fixed set of N items with human utilities $u_m(i)$ and means μ_m. An audit labels a pilot of n_{pilot} items fully, picks the candidate $\hat{m}$ with the best pilot mean, and *certifies* it if for every competitor j a one-sided upper bound on $\mu_j - \mu_{\hat{m}}$ is at most ε (normal bound, Bonferroni over the $|\mathcal{M}| - 1$ competitors, $\alpha = 0.10$). A certificate is wrong if $\max_j \mu_j - \mu_{\hat{m}} > \varepsilon$. Menus are deliberately hard, built from the full human labels (the top-4 systems by MQM, every pair among the top 6, the closest Arena pairs); their construction is not part of the audit.

Estimator. After the pilot, items are drawn by Poisson sampling with probabilities π_i. For $D_j = u_j - u_{\hat{m}}$ and an evaluator's predicted difference $\hat{D}_j$,

$$\hat{\Delta}_j = \tfrac{1}{N}\Big[\sum_{\text{pilot}} D_j + \sum_{\text{rest}} \lambda_j \hat{D}_j + \sum_{i \in S} \tfrac{D_j(i) - \lambda_j \hat{D}_j(i)}{\pi_i}\Big], \tag{1}$$

with $\lambda_j = \max(0, \widehat{\text{Cov}}(D_j, \hat{D}_j)/\widehat{\text{Var}}(\hat{D}_j))$ the PPI++ coefficient, either fitted on the pilot and frozen or refitted on every label as the audit proceeds; $\lambda = 0$ is the human-only estimator; where two outputs coincide, $\hat{D}_j(i)$ is set to 0 with $D_j(i) = 0$ (the audit code does this explicitly; 13–14 of the WMT22 and 26–27 of the WMT23 metrics score identical strings slightly differently, by at most 2×10^{-4} in the mean difference otherwise). With a frozen coefficient, (1) is a Horvitz–Thompson estimator (Horvitz and Thompson, 1952) and, given the pilot, unbiased for the finite-population mean for any evaluator; its variance is estimated by $\sum_S (1 - \pi_i) R_i^2/(\pi_i^2 N^2)$, $R = D - \lambda \hat{D}$. With the coefficient refitted on the sampled labels themselves, λ depends on the sample, exact unbiasedness no longer holds and the variance estimate ignores the coefficient's error, the plug-in bias Chaganty et al. (2018) note; we therefore report the coverage of the refitted bound by simulation (§3.3, App. D).

Designs. *Uniform*: constant π, every menu output of a sampled item labelled, identical strings included, as in WMT22's own MQM campaign (42% of en→de and 18% of zh→en rated outputs share their string with another system's). *Dedup*: uniform over the items where some competitor's output differs from the candidate's, identical strings labelled once, and $D_j = 0$ used exactly where outputs coincide. A *weighted* variant that also samples items in proportion to a label-free proxy of $|D|$ is sometimes cheaper and is reported in App. E; the cheapest of these human-only designs in a cell is its *judge-free baseline*, a benchmark chosen from realised costs. An evaluator enters only as the control variate of (1) on top of a design.

Cost and HES. J_{50} counts unique human labels, pilot included, at which a pre-committed budget certifies in half of the simulated audits (300 or 2,000; linear interpolation over a geometric grid of 14 budgets). The human-evidence saving of an evaluator is $\mathrm{HES} = 1 - J_{50}(\text{design} + \text{evaluator})/J_{50}(\text{design})$. The pilot's cost B_{pilot} is its unique labels (one per distinct output in MT): 162 (en→de) and 182 (zh→en) for the 50-segment MT pilot, 42–68% of J_{50} on en→de; an evaluator can act only on the rest, the factor $1 - B_{\text{pilot}}/J$ of §4. Intervals are paired bootstraps over the simulated audits of a fixed test set, with one resampling of audit indices shared by every cell of a comparison (the cells share the simulator's random streams): Monte-Carlo, not data-sampling, uncertainty.

Data and evaluators. MT: WMT22 MQM (Freitag et al., 2021, 2022) en→de (1,315 segments) and zh→en (1,875), Google weights, $u = -\min(\mathrm{MQM}, 25)/25$, identical strings of a segment sharing the mean of their ratings; the 31 WMT22 metric variants available at segment level in mt-metrics-eval (reference-based against refA, including COMET-22 (Rei et al., 2022), and reference-free QE variants), plus our own COMET-22 run and Qwen3-8B (Yang et al., 2025) and Mistral-7B-Instruct-v0.3 (Jiang et al., 2023) with the GEMBA-DA prompt (Kocmi and Federmann, 2023b); pilot 50, $\varepsilon \in \{0.01, 0.02\}$. The held-out year is WMT23 MQM en→de (460 paragraph-level segments, 11 systems) and zh→en (1,177 segments, 14 systems; Freitag et al., 2023) with its 42 submitted metric variants, among them XCOMET (Guerreiro et al., 2024), MetricX-23 (Juraska et al., 2023) and the GPT-4-based GEMBA-MQM (Kocmi and Federmann, 2023a) (App. H). **Chat:** the six English LMArena pairs (Chiang et al., 2024) with the smallest win-rate gaps among those with ≥ 300 battles (gaps 0.011–0.084); Qwen3-8B and Mistral-7B as pairwise judges, both presentation orders averaged, and "the longer response wins"; pilot 50, $\varepsilon \in \{0.05, 0.1\}$. **Retrieval:** a retrospective re-analysis of fixed-budget audit records generated in earlier, unpublished experiments (setup in App. A): three cutoff policies compared by precision at their cutoffs on four fully judged collections (ANTIQUE, CAsT 2019, DBpedia-Entity, TREC DL 21–23; Hashemi et al., 2020; Dalton et al., 2020; Hasibi et al., 2017; Craswell et al., 2021, 2022, 2023) with Qwen3-8B (UMBRELA prompt; Upadhyay et al., 2024) and Qwen3-Reranker-0.6B (Zhang et al., 2025); pilot 20 queries.

3 Results

3.1 The design takes most of the saving

Figure 1 splits the saving over uniform sampling into the part the judge-free design obtains and the part an evaluator adds on the same design, in the same audits: with s_D the design's share and h the evaluator's HES over it, the total is $s_{\text{total}} = s_D + (1 - s_D)h$, so the two are never added directly. In MT, identical outputs (per pair of menu systems, 14–24% of en→de and 5–16% of zh→en segments) need no label and many paired MQM differences are exactly zero (53–59% and 40–48%); skipping them saves a median 5.2% (WMT22) and 3.8% (WMT23) of uniform-sampling labels over the two-system decisions, and 8–16% in the four pre-registered top-4 cells (App. A, Table 2). In retrieval, where whole documents fall outside every cutoff, the design saves 41–56% and the Qwen3-8B judge adds 1–9 points.

The average evaluator adds little. On the dedup design the mean of the submitted metrics adds a median 1.8% (WMT22) and 2.5% (WMT23) with a refitted coefficient, and nothing (0.2%, 0.3%) with a coefficient frozen after the pilot; on uniform sampling the same evaluators would add 2.1% and 2.9%, so the design takes little of what they have to give, and in 69–79% of cells it saves more labels than the mean evaluator adds. The pre-fixed COMET-22 adds 2.9% and 3.1%.

Strong evaluators are the exception. The strongest metric of each year, MetricX-XXL on WMT22 and XCOMET-XXL on WMT23, adds

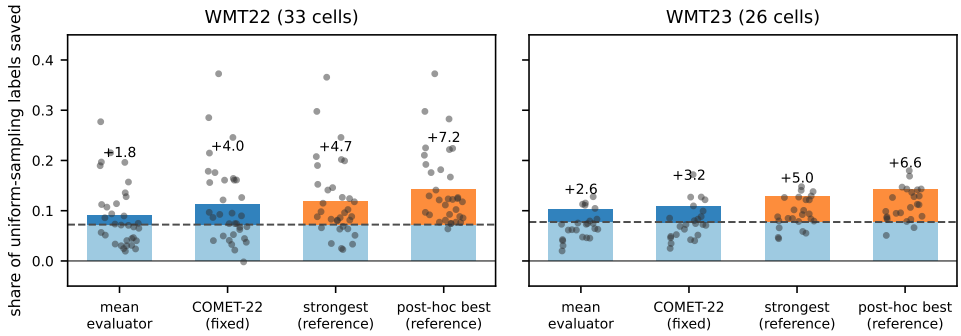


Figure 1: Same-condition decomposition of the saving over uniform sampling (same menus, pilot, ε and simulated audits). Per cell, $s_{\text{total}} = 1 - D/A = s_D + (1 - s_D)h$ with $s_D = 1 - C/A$ the judge-free dedup design's share (light) and $(1 - s_D)h$, $h = 1 - D/C$, the evaluator's on the same design with a refitted coefficient (dark); the bars stack the means over informative cells of the two parts, so their sum is the mean total (medians, which the text reports, do not add). Mean evaluator and the pre-fixed COMET-22 are strategies an auditor can run; the strongest evaluator of the year (MetricX-XXL, XCOMET-XXL) and the post-hoc best are chosen from the outcome and shown as reference. Dots: cells. WMT22: 33 cells, 300 audits, 31 metrics; WMT23: 26 cells, 2,000 audits, 42 metrics.

a median 3.7% and 4.7% refitted, and the post-hoc best 6.8% and 6.2%. In labels, the design saves more than XCOMET-XXL adds in only 11 of the 26 WMT23 cells (22 of 33 for MetricX-XXL on WMT22), and more than the ten evaluators with the highest pilot ρ in 44% (WMT23) and 67% (WMT22) of decision–evaluator pairs. The statement "the design saves more than the evaluator" therefore holds for the typical evaluator and for every evaluator with a frozen coefficient, not for the strongest with a refitted one; whether the strongest can be identified in advance from a pilot is the question of §3.3 (the 50-segment pilot-ρ rule we tested does not). On the six close chat pairs the open 7–8B judges reach decision-level ρ of -0.01 to 0.29 and save at least 5% in two pairs (H6 required three; App. I).

3.2 Thirty decisions: small, real, bounded

Four cells are few. We therefore repeated the comparison on every two-system decision among the top six WMT22 systems of each language pair: 30 decisions, two ε each, the 31 metric variants, 2,000 simulated audits per cell (the median 95% interval of one evaluator's HES is then ± 2.3 points; the 300-audit runs with all 34 evaluators give the same medians, App. E). In 36 of the 60 cells at least 30% of the uniform-sampling cost falls after the pilot; in the others the pilot nearly settles the decision and nothing can save much. Over these 36 cells (Figure 2) the judge-free design saves a median 4.6%; the mean evaluator adds 0.2% with the frozen coefficient and 2.3% refitted, the ten strongest 3.9%, and the design saves more than the mean evaluator in 100% and 89% of cells. The post-hoc best (median 6.0% refitted) leads the mean by more than selection noise explains (3.3 vs. 1.8 points; $p < 0.05$ in 72% of cells), so evaluators do differ; the design still beats it in 39% of cells. The one-sided 95% upper bound of a single evaluator's refitted HES is below 5% in 64% and below 10% in 94% of the 1,116 decision–evaluator pairs (29% and 88% for the ten strongest); 53% of pairs are significantly above zero, with a median 2.0%. In labels: of a median decision's 256, the design saves 17, the mean evaluator 6, the post-hoc best 17.

3.3 Held-out year: what an auditor should do

Everything above chooses the cheapest design and the best evaluator after seeing the costs. An auditor cannot. We therefore fixed four strategies on WMT22 and applied them, unchanged, to the WMT23 MQM sets (the same 15 two-system menus per language pair, pilot 50, $\varepsilon \in \{0.01, 0.02\}$, 42 metric variants; pre-registered as lock v1.2 before any WMT23 record was examined, App. H). S1: the dedup design alone (cheaper than weighted in 75–83% of WMT22 cells). S2: dedup plus COMET-22, the WMT22 metric with the highest decision-level ρ among those submitted under the same name in both years. S3: dedup plus the evaluator with the highest pilot ρ. S4: S3 unless that ρ is below 0.2, the break-even of §4; then S1. Each strategy picks one arm per audit from pilot

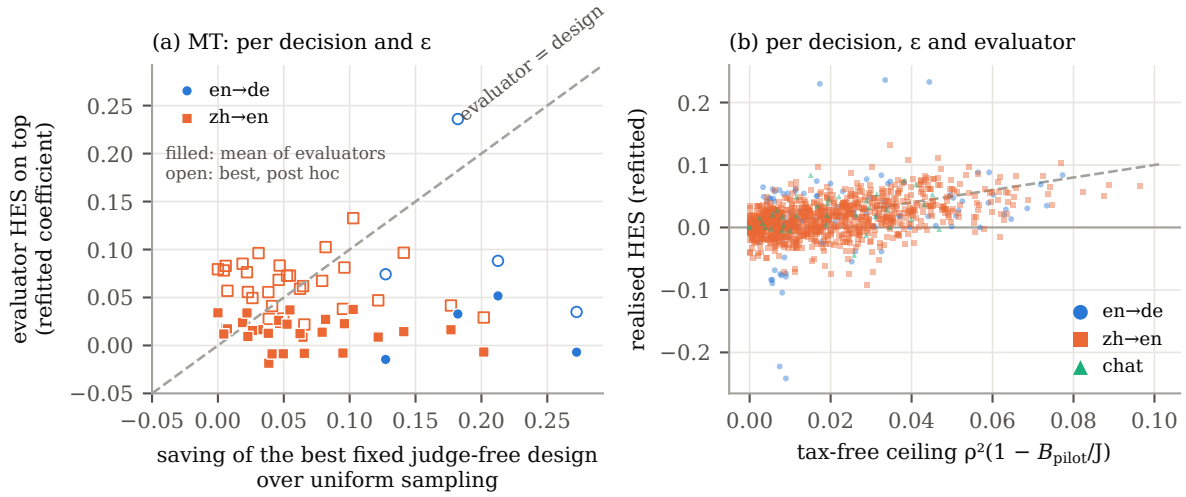


Figure 2: The 30 two-system WMT22 decisions at 300 audits per cell (the text of §3.2 reports the 2,000-audit re-run, 36 cells, whose medians agree). (a) The 33 MT decision–ε cells (19 decisions) in which at least 30% of the uniform-sampling cost falls after the pilot: saving of the best fixed judge-free design over uniform sampling against the HES of the evaluators on top of it (refitted coefficient; filled: mean of the 34 evaluators, open: best, post hoc). (b) Every informative decision, ε and evaluator, MT and chat (1,158 points): realised HES against the tax-free approximation $\rho^2(1 - B_{\text{pilot}}/J)$ of (2), with the pilot-median ρ.

	refitted		frozen	
cost relative to S1 (dedup)	WMT22	WMT23	WMT22	WMT23
informative cells	33	26	33	26
S2 fixed COMET-22	−4.4 (85)	−3.4 (100)	−2.2 (76)	−0.4 (65)
S3 pilot selection	−3.2 (88)	−3.3 (85)	+2.3 (27)	+4.2 (23)
S4 select or abstain	−3.2 (88)	−3.2 (85)	+2.2 (27)	+4.2 (23)
post-hoc best (reference)	−7.7	−7.0	−6.2	−3.5
mean evaluator (reference)	−2.0	−2.8	+0.1	−0.0
S3 costlier than S1 by $> 2\%$	3%	8%	45%	50%
wrong certificates at J_{50}, S1 / S3	.003 / .003	.007 / .006	.003 / .003	.007 / .005

Table 1: Mean excess cost over the judge-free dedup design, in % of its J_{50} (negative: cheaper), with the share of informative cells in which the strategy is cheaper in parentheses; WMT22 at 300 simulated audits per cell, WMT23 at 2,000; unrestricted protocol (the certificate taken at face value at every budget). Design saving over uniform: median 5.2% (WMT22) and 3.8% (WMT23); pilot share 37% and 54%. Decision-level ρ: WMT22 at most 0.28 over the 31 metrics (COMET-22 rank 4), WMT23 at most 0.36 over 42 (MetricX-23-QE, XCOMET; COMET-22 rank 21; GEMBA-MQM 0.22). The post-hoc best leads the mean evaluator by more than selection noise in 81% of the WMT23 cells.

information only; its cost is J_{50} with the pilot included. Table 1 gives the outcome, and three paired comparisons answer three questions.

Is an evaluator worth adding? S2 against S1: with the coefficient refitted, the pre-fixed COMET-22 costs 3.4% less than the design alone on WMT23 (95% Monte-Carlo interval [2.9, 3.9]; cheaper in all 26 informative cells, 21 of them with a per-cell interval above zero, and in 31 of the 60 cells overall, the other 29 being decisions settled at the pilot where every strategy costs the same; WMT22: 4.4% [2.1, 6.7], 85% of cells). With the coefficient frozen after the pilot it saves 0.4% (WMT22: 2.2%).

Is choosing the evaluator from the pilot worth more than fixing one in advance? S3 against S2 on the same audits, $G = 1 - J_{50}(\text{S3})/J_{50}(\text{S2})$: -0.1% on WMT23 (95% Monte-Carlo interval $[-0.7, +0.5]$: the mean over cells, with one resampling of the 2,000 audit indices applied to every cell at once, since the cells share the simulator's random streams; the benchmark itself is fixed) and -1.5% on WMT22 ($[-4.4, +0.3]$, 300 audits). Treating the decisions as clusters widens the intervals to $[-1.2, +1.1]$ and $[-8.9, +1.3]$ (App. H). Neither year shows a gain from selection, and every interval lies below $+2$ points, the margin we set after the 300-audit results and before the 2,000-audit run (it is not part of lock v1.2): a one-sided check that selection does not gain 2 points, not a two-sided equivalence test, and on WMT22 a loss of several points remains open. The 60 WMT23 cells give -0.1% $[-0.3, +0.2]$. Abstaining when the chosen evaluator's pilot ρ is below 0.2 changes nothing (2–4% of audits). The post-hoc best is 7.0% cheaper than S1 and is shown for reference only: it is chosen from the outcome. At 2,000 audits its lead over the *mean* evaluator exceeds what selecting the largest of 42 equal evaluators produces in 81% of

cells (the test of §3.2, not a test against S2; at 300 audits in none), so the evaluators do differ; the pilot selection we tested (the highest ρ on 50 segments) does not find the best one, and adds nothing over the pre-fixed COMET-22.

Does the coefficient rule matter? S3 frozen against S3 refitted: pilot selection costs 4.2% *more* than the design with a frozen coefficient and 3.3% less with a refitted one (WMT22: $+2.3\%$ and -3.2%); half of the frozen-coefficient selections are over 2% dearer than the design, 8% of the refitted ones. The price of a pilot coefficient is the evaluator's whole value.

Validity. On the 30 decisions themselves, whose true gaps are mostly well clear of ε, every strategy certifies at its J_{50} with a wrong-certificate rate of 0.003–0.006, equal to the human-only design's, and the chosen arm's upper bound covers the true difference at 0.89–0.90 at that budget (nominal 0.90), at 0.90–0.91 averaged over all budgets and at 0.86–0.89 at the worst budget of a cell, the same as the human-only design (0.86–0.89). This holds once known-zero differences are set to zero (§2); on records that keep the small non-zero scores 26–27 of the WMT23 metrics give identical outputs, the refitted bound covered at only 0.45–0.50 at the smallest budgets: the term $\lambda \sum_{\text{rest}} \hat{D}$ of (1) is then biased on the known-zero items, and a coefficient fitted on few labels amplifies it (App. H). Whether the certificate keeps its nominal error where it matters is a different question, answered under boundary stress (the runner-up as candidate, ε at 0.9 times its regret, 300 audits), where every certificate is wrong. At the smallest post-pilot budget (ten sampled items, about 20 labels on top of the pilot's 95) the type-I error is 0.18 for the human-only design, 0.17 for S2 and 0.12 for S3; the observed mean rate is at or below the nominal 0.10 at every later budget from about 206 (en→de) and 265 (zh→en) labels in total, for S1 and S2 alike, and at or below 0.125 from about 145 labels (App. H). The evaluator strategies are below the human-only design on average (by 0.1–1.5 points) and above it at 15–34% of the cell–budget points, by more than 2 points at 1–3%. The inflation is the normal bound on few items; it applies to the human-only baseline as much as to the evaluators, so it does not restore the certificate, and it is an observation on this simulation rather than a validated range. As a post-hoc sensitivity analysis we therefore re-ran the comparison under a protocol that issues no certificate before those budgets, the same rule for every strategy, on the same 26 cells: with the 0.10 cut-off (55 and 88 post-pilot items) 22 of the 26 cells certify at the first allowed budget whatever the strategy and S2's saving over S1 falls from 3.4% to 0.4% [0.2, 0.6] (0.2% over all 60 cells); with the 0.125 cut-off (23 and 29 items) it is 3.0% [2.5, 3.4] (1.3% over 60). The 3% of Table 1 is therefore the saving of an auditor who takes the certificate at face value at 140–200 labels, where its boundary type-I error is 0.11–0.12 rather than 0.10; an auditor who waits for the 0.10 range keeps almost none of it, because most of these 30 decisions certify earlier, and the ranking of the strategies is the same under every rule. The new judges raise decision-level ρ from 0.28 to 0.36 (MetricX-23, XCOMET; GEMBA-MQM 0.22). The rule that survives both years is short: deduplicate; if the decision is expected to need more labels than the pilot and the first hundred or so after it, add one strong evaluator fixed in advance and refit its coefficient; do not spend the pilot on choosing the evaluator or on committing to the coefficient. For a decision that settles within those minimum budgets, no evaluator has room to save much.

4 Why: the level of measurement and the cost of the pilot

Evaluator quality collapses across levels. Among the 11 WMT22 submissions the strong metrics correlate with MQM at 0.93–0.99 at system level. Among the four closest systems the ranking is fragile: MetricX-XL-DA and MetricX-XXL-DA reach system-level Pearson 0.994–0.995 on zh→en and Kendall $\tau = 0$ among the top four, and 10 of the 31 zh→en variants (system-level 0.83–0.96; BLEU, spBLEU, chrF, BERTScore, YiSi, SEScore, MEE) rank them in exactly reversed order. The quantity that governs a control variate is the correlation ρ of per-item paired differences, and it is low everywhere: 0.30–0.38 for the best metrics on en→de, 0.21–0.25 on zh→en, 0.17–0.29 for COMET-22 and the Qwen3-8B judge, 0.04–0.09 for Mistral-7B (Figure 4 in App. A). Part of this is label noise: in the en→de three-rater re-annotation two raters' paired differences correlate only 0.31–0.41, so even a perfect evaluator would reach $\rho \approx 0.59$ against the single ratings an audit collects; the best metrics attain one half to two thirds of that (App. E). HES is ranked slightly better by ρ than by system-level Pearson (Spearman 0.39 vs. 0.36 on en→de, 0.22 vs. 0.07 on zh→en;

H11), but neither predicts it well, because every real evaluator sits where the saving is small.

The saving is about $\rho^2(1 - B_{\text{pilot}}/J)$. A control variate with correlation ρ reduces the residual variance by the factor $1 - \rho^2$ (Angelopoulos et al., 2023b); with λ estimated on a pilot of P_{eff} informative items and frozen, the factor becomes $(1-\rho^2)(1+1/P_{\text{eff}})$ (Cochran, 1977); and only the post-pilot share $1 - B_{\text{pilot}}/J_0$ of the cost can be saved. To first order,

$$\text{HES} \approx \left[\rho^2 - \frac{1-\rho^2}{P_{\text{eff}}}\right]\left(1 - \frac{B_{\text{pilot}}}{J_0}\right), \quad (2)$$

with $J_0 = B_{\text{pilot}} + L_0$, an interpretive approximation (homoskedastic linear projection, normal fixed-budget approximation; derivation and the design-optimal coefficient in App. B), not a bound. It says three things that the data confirm. A frozen coefficient costs labels when $\rho^2 < 1/(P_{\text{eff}} + 1)$, about $\rho < 0.2$ for the 50-item MT pilot where half the paired differences are zero; this is why pilot selection loses in §3.3 and why the frozen-coefficient "tax" is 4.3 [3.5, 5.2] (en→de) and 2.5 [2.0, 3.1] (zh→en) points over the 31 metrics (H9), removed by refitting on all labels (coverage of the upper bound 0.90 at nominal 0.90; App. C, D). Refitted savings follow the tax-free approximation $\rho^2(1 - B_{\text{pilot}}/J)$ (median deviation -0.4 points over 1,158 decision–evaluator points), and it is below 5% for 94% of them, because $\rho \leq 0.43$ and the pilot is a median 39% of the cost. Under controlled variation (semi-synthetic evaluators with ρ set to 0.1–0.9, 126 cells) predicted and realised HES correlate 0.97, with 10% realised at $\rho = 0.5$ and 31% at $\rho = 0.8$ (Figure 5, App. G). The pilot size does not change the picture: pilots of 25 and 10 segments give the mean evaluator 0.7% and 0.3% refitted and -2.7% and -7.7% frozen, against design savings of 4.3–4.4% (App. E).

When the answer changes. The three factors of (2) say where these results do and do not transfer. When decision-level ρ is low, as for every WMT22 metric and most WMT23 ones, the increment is a few percent whatever the design; the exception in §3.1 is exactly the evaluators with ρ above 0.3. When the pilot is a large share of the cost, as on the paragraph-level WMT23 en→de set (53%) or whenever the decision is easy, little remains for any evaluator to save, which is why we report only cells with at least 30% of the cost after the pilot. When a strong evaluator is used with a refitted coefficient, the increment can match the design's own saving; with a frozen coefficient it cannot, because the tax $(1-\rho^2)/P_{\text{eff}}$ consumes it. What we count is human labels: the evaluator's own cost, annotation time per label and the price of organising deduplication are outside the measure. A label-cost model would re-weight the bars of Figure 1; we did not measure those costs, so the ordering by total cost is not established here.

5 Related work

The unit of evaluation. System-level correlation is driven by outliers (Mathur et al., 2020), close systems are better compared by pairwise accuracy (Kocmi et al., 2021; Deutsch et al., 2023), correlations on close pairs are much lower (Deutsch et al., 2022), and such comparisons are often underpowered (Card et al., 2020). JuStRank (Gera et al., 2025) and Gao et al. (2025) show that instance-level accuracy does not determine system-ranking quality and that ranking degrades for close systems; Zhang et al. (2026) separate judgment quality from downstream utility in training and inference. We take the unit one step further, to the human labels a close decision requires.

Saving human labels with an evaluator. Chaganty et al. (2018) asked our question first: using automatic metrics as control variates on human judgments of summarisation and question answering, they obtained savings of 7–13%, showed that a metric's high system-level correlation does not carry over to the instance level where the control variate acts, that human rating noise caps the attainable saving, and that a coefficient estimated on the same data is slightly biased. PPI and PPI++ (Angelopoulos et al., 2023a,b) turned this into a general estimator; AutoEval (Boyeau et al., 2025), prediction-powered ranking (Chatzi et al., 2024), noisy-judge inference (Chen et al., 2026), PRECISE (Divekar, 2026) and Gligorić et al. (2025) apply it to LLM evaluation, and Dorner et al. (2025) bound the saving of an evaluator no better than the evaluated model. Closest to us, Gao et al. (2026) rank WMT metrics by the annotation they save under prediction-powered evaluation (PPSR, the mean squared correlation of paired differences), test system pairs, and verify the saving by resampling at 80% power under uniform labelling with the coefficient fitted on the evaluation labels. Their saving and ours are not directly comparable (power 80% vs. our J_{50}; uniform vs. dedup; all pairs vs.

the closest), which is why Figure 1 measures the evaluator on both designs in the same audits.

Choosing what humans label. Which items humans label is studied for model comparison (Sawade et al., 2012), active testing (Kossen et al., 2021) and IR test collections (Sakai, 2016; Li and Kanoulas, 2017; Oosterhuis et al., 2024); within prediction-powered inference, active inference (Zrnic and Candès, 2024a) allocates labels by the evaluator's uncertainty, stratified PPI (Fisch et al., 2024) fits the coefficient per stratum, and cross-prediction (Zrnic and Candès, 2024b) splits the labels between fitting the predictor and inference (our cross-fitted coefficient borrows the split, not the method). These works show that the design and the evaluator both set the cost; the evaluator-driven designs are outside what we measure.

What this paper adds. The increment of an evaluator used as a control variate once a judge-free design has been credited and the pilot paid, measured as the labels a close decision needs; the finding that this increment is small for the typical evaluator and comparable to the design's own saving for the strongest; and a test, with rules fixed on one year and applied to the next, of whether an auditor can obtain it without choosing the evaluator after the fact.

6 Conclusion: three numbers to check before an audit

For a close system decision, an evaluator's value is set by the correlation of the differences the decision depends on, diluted by the pilot's fixed cost and by what a judge-free design already obtains; WMT22 and WMT23 metrics then save a few percent of human labels on average, the strongest somewhat more, and the typical one less than organising the annotation itself. A team can check three numbers before paying for an evaluator: (i) its decision-level ρ on the systems at stake, from a prior campaign or the pilot, not its system-level agreement; (ii) the pilot's share of the cost, B_{pilot}/J, which a human-only pilot prediction supplies; (iii) what the judge-free design already saves: its duplicate and known-zero structure is known in advance, the J_{50} it saves has to be estimated, as here by simulation. Under (2) the increment is about $\rho^2(1 - B_{\text{pilot}}/J)$: for the WMT metrics ($\rho \leq 0.43$, $B_{\text{pilot}}/J \approx 0.4$) up to about 11%, typically 2–5%, against design savings of 5–16%. The strategy these numbers imply, deduplicate, add one strong evaluator fixed in advance, refit its coefficient, saved 3% over the design on a held-out year when the certificate is taken at face value, and 0.4% (3.0% at a tolerated error of 0.125) under a protocol that certifies only where the bound's observed error is at nominal, since most decisions settle earlier; choosing the evaluator from the pilot instead was not measurably better, and freezing the coefficient cost the evaluator's whole value.

Limitations

The menus are built from full human labels to be deliberately hard; easier decisions leave less to save for any method. The certificate uses a normal bound whose calibration we checked by simulation, not proved; under boundary stress its type-I error is above nominal below the minimum budgets read off the two WMT23 language pairs (about 206 and 265 labels), for every strategy, and the savings reported at face value include decisions that certify in that range; the budget cut-offs were read off the same boundary simulation and checked on a second seed, but they are observations, not a calibrated bound (§3.3). MT savings are computed on the MQM-rated subsets of WMT22 (two language pairs), chat on six pairs from a 2023 Arena release, retrieval on a retrospective re-analysis of audit records from earlier, unpublished experiments with two judges. The cost approximation is first-order; it describes controlled variation of ρ and the scale of real savings, but does not predict them cell by cell. The LLM judges we ran are open 7–8B models; WMT23 contributes the submitted scores of the GPT-4-based GEMBA-MQM (ρ 0.22), but we ran no new inference with frontier models, whose decision-level ρ may be higher. The evaluator enters only as a control variate on the audit; its other uses (choosing the candidate, building the menu, stratifying the sample) are outside the decision value we measure. HES depends on the pilot size, and larger pilots lower the estimation tax while raising the fixed cost (§4). The results should not be read as showing that automatic evaluators are useless: they concern close decisions, small human pilots and evaluators with decision-level $\rho \leq 0.43$; by the cost law, an evaluator with $\rho \approx 0.8$ would save about a third of the labels with our pilots, and more with a smaller pilot share, but against single WMT22 ratings label noise caps ρ near 0.6 (App. E), so such savings also require

more reliable human labels. Uncertainty is Monte-Carlo over simulated audits of one fixed test set per language pair; items and raters are not resampled. We report J_{50}; at 80% power savings differ and are often censored by the budget grid (Appendix D). The estimand treats identical strings as one quality; sharing one random rating per string instead leaves the design savings essentially unchanged (App. E), but under an as-rated estimand, in which identical strings may differ in quality, their rating noise would be variance that deduplication does not remove. The retrieval analysis reuses audit records from earlier, unpublished experiments rather than an independently collected evaluation campaign. Its findings are limited to the evaluated retrieval policies, collections, and judges.

References

Anastasios N. Angelopoulos, Stephen Bates, Clara Fannjiang, Michael I. Jordan, and Tijana Zrnic. 2023a. Prediction-powered inference. *Science*, 382(6671):669–674.

Anastasios N. Angelopoulos, John C. Duchi, and Tijana Zrnic. 2023b. PPI++: Efficient prediction-powered inference. *arXiv preprint arXiv:2311.01453*.

Alexander Bondarenko, Maik Fröbe, Meriem Beloucif, Lukas Gienapp, Yamen Ajjour, Alexander Panchenko, Chris Biemann, Benno Stein, Henning Wachsmuth, Martin Potthast, and Matthias Hagen. 2020. Overview of Touché 2020: Argument retrieval. In *Experimental IR Meets Multilinguality, Multimodality, and Interaction: 11th International Conference of the CLEF Association (CLEF 2020)*, volume 12260 of *Lecture Notes in Computer Science*, pages 384–395. Springer.

Pierre Boyeau, Anastasios Nikolas Angelopoulos, Tianle Li, Nir Yosef, Jitendra Malik, and Michael I. Jordan. 2025. Autoeval done right: Using synthetic data for model evaluation. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 5276–5290. PMLR.

Dallas Card, Peter Henderson, Urvashi Khandelwal, Robin Jia, Kyle Mahowald, and Dan Jurafsky. 2020. With little power comes great responsibility. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*, pages 9263–9274.

Arun Chaganty, Stephen Mussmann, and Percy Liang. 2018. The price of debiasing automatic metrics in natural language evaluation. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 643–653.

Ivi Chatzi, Eleni Straitouri, Suhas Thejaswi, and Manuel Gomez-Rodriguez. 2024. Prediction-powered ranking of large language models. In *Advances in Neural Information Processing Systems*.

Yiqun T. Chen, Sizhu Lu, Sijia Li, Moran Guo, and Shengyi Li. 2026. Efficient inference for noisy LLM-as-a-judge evaluation. In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *PMLR*, pages 15836–15860.

Wei-Lin Chiang, Lianmin Zheng, Ying Sheng, Anastasios Nikolas Angelopoulos, Tianle Li, Dacheng Li, Hao Zhang, Banghua Zhu, Michael Jordan, Joseph E. Gonzalez, and Ion Stoica. 2024. Chatbot arena: An open platform for evaluating LLMs by human preference. In *Proceedings of the 41st International Conference on Machine Learning*.

William G. Cochran. 1977. *Sampling Techniques*, 3rd edition. Wiley.

Nick Craswell, Bhaskar Mitra, Emine Yilmaz, Daniel Campos, and Jimmy Lin. 2021. Overview of the TREC 2021 deep learning track. In *Proceedings of the Thirtieth Text REtrieval Conference (TREC 2021)*. ArXiv:2507.08191.

Nick Craswell, Bhaskar Mitra, Emine Yilmaz, Daniel Campos, Jimmy Lin, Ellen M. Voorhees, and Ian Soboroff. 2022. Overview of the TREC 2022 deep learning track. In *Proceedings of the Thirty-First Text REtrieval Conference (TREC 2022)*. ArXiv:2507.10865.

Nick Craswell, Bhaskar Mitra, Emine Yilmaz, Hossein A. Rahmani, Daniel Campos, Jimmy Lin, Ellen M. Voorhees, and Ian Soboroff. 2023. Overview of the TREC 2023 deep learning track. In *Proceedings of the Thirty-Second Text REtrieval Conference (TREC 2023)*. ArXiv:2507.08890.

Jeffrey Dalton, Chenyan Xiong, and Jamie Callan. 2020. TREC CAsT 2019: The conversational assistance track overview. *arXiv preprint arXiv:2003.13624*.

Daniel Deutsch, Rotem Dror, and Dan Roth. 2022. Re-examining system-level correlations of automatic summarization evaluation metrics. In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 6038–6052.

Daniel Deutsch, George Foster, and Markus Freitag. 2023. Ties matter: Meta-evaluating modern metrics with pairwise accuracy and tie calibration. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 12914–12929.

Abhishek Divekar. 2026. Statistically reliable LLM-based ranking evaluation via prediction-powered inference. *arXiv preprint arXiv:2606.05308*.

Florian E. Dorner, Vivian Y. Nastl, and Moritz Hardt. 2025. Limits to scalable evaluation at the frontier: LLM as judge won't beat twice the data. In *International Conference on Learning Representations*.

Adam Fisch, Joshua Maynez, R. Alex Hofer, Bhuwan Dhingra, Amir Globerson, and William W. Cohen. 2024. Stratified prediction-powered inference for effective hybrid evaluation of language models. In *Advances in Neural Information Processing Systems*.

Markus Freitag, George Foster, David Grangier, Viresh Ratnakar, Qijun Tan, and Wolfgang Macherey. 2021. Experts, errors, and context: A large-scale study of human evaluation for machine translation. *Transactions of the Association for Computational Linguistics*, 9:1460–1474.

Markus Freitag, Nitika Mathur, Chi-kiu Lo, Eleftherios Avramidis, Ricardo Rei, Brian Thompson, Tom Kocmi, Frederic Blain, Daniel Deutsch, Craig Stewart, Chrysoula Zerva, Sheila Castilho, Alon Lavie, and George Foster. 2023. Results of WMT23 metrics shared task: Metrics might be guilty but references are not innocent. In *Proceedings of the Eighth Conference on Machine Translation*, pages 578–628.

Markus Freitag, Ricardo Rei, Nitika Mathur, Chi-kiu Lo, Craig Stewart, Eleftherios Avramidis, Tom Kocmi, George Foster, Alon Lavie, and André F. T. Martins. 2022. Results of WMT22 metrics shared task: Stop using BLEU – neural metrics are better and more robust. In *Proceedings of the Seventh Conference on Machine Translation (WMT)*, pages 46–68.

Mingqi Gao, Yixin Liu, Xinyu Hu, Xiaojun Wan, Jonathan Bragg, and Arman Cohan. 2025. Re-evaluating automatic LLM system ranking for alignment with human preference. In *Findings of the Association for Computational Linguistics: NAACL 2025*.

Mingqi Gao, Anthony Sicilia, and Weiyan Shi. 2026. Which metrics save the most human annotation? prediction-powered evaluation and meta-evaluation. In *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing*. To appear; arXiv:2608.26638.

Ariel Gera, Odellia Boni, Yotam Perlitz, Roy Bar-Haim, Lilach Eden, and Asaf Yehudai. 2025. JuStRank: Benchmarking LLM judges for system ranking. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*.

Kristina Gligorić, Tijana Zrnic, Cinoo Lee, Emmanuel Candès, and Dan Jurafsky. 2025. Can unconfident LLM annotations be used for confident conclusions? In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 3514–3533.

Nuno M. Guerreiro, Ricardo Rei, Daan van Stigt, Luisa Coheur, Pierre Colombo, and André F. T. Martins. 2024. xCOMET: Transparent machine translation evaluation through fine-grained error detection. *Transactions of the Association for Computational Linguistics*, 12:979–995.

Helia Hashemi, Mohammad Aliannejadi, Hamed Zamani, and W. Bruce Croft. 2020. ANTIQUE: A non-factoid question answering benchmark. In *European Conference on Information Retrieval*.

Faegheh Hasibi, Fedor Nikolaev, Chenyan Xiong, Krisztian Balog, Svein Erik Bratsberg, Alexander Kotov, and Jamie Callan. 2017. DBpedia-Entity v2: A test collection for entity search. In *Proceedings of the 40th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 1265–1268.

Daniel G. Horvitz and Donovan J. Thompson. 1952. A generalization of sampling without replacement from a finite universe. *Journal of the American Statistical Association*, 47(260):663–685.

Albert Q. Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, Lélio Renard Lavaud, Marie-Anne Lachaux, Pierre Stock, Teven Le Scao, Thibaut Lavril, Thomas Wang, Timothée Lacroix, and William El Sayed. 2023. Mistral 7B. *arXiv preprint arXiv:2310.06825*.

Juraj Juraska, Mara Finkelstein, Daniel Deutsch, Aditya Siddhant, Mehdi Mirzazadeh, and Markus Freitag. 2023. MetricX-23: The Google submission to the WMT 2023 metrics shared task. In *Proceedings of the Eighth Conference on Machine Translation*, pages 756–767.

Tom Kocmi and Christian Federmann. 2023a. GEMBA-MQM: Detecting translation quality error spans with GPT-4. In *Proceedings of the Eighth Conference on Machine Translation*, pages 768–775.

Tom Kocmi and Christian Federmann. 2023b. Large language models are state-of-the-art evaluators of translation quality. In *Proceedings of the 24th Annual Conference of the European Association for Machine Translation*, pages 193–203.

Tom Kocmi, Christian Federmann, Roman Grundkiewicz, Marcin Junczys-Dowmunt, Hitokazu Matsushita, and Arul Menezes. 2021. To ship or not to ship: An extensive evaluation of automatic metrics for machine translation. In *Proceedings of the Sixth Conference on Machine Translation*, pages 478–494.

Jannik Kossen, Sebastian Farquhar, Yarin Gal, and Tom Rainforth. 2021. Active testing: Sample-efficient model evaluation. In *International Conference on Machine Learning*.

Dawn Lawrie, James Mayfield, Eugene Yang, Andrew Yates, Sean MacAvaney, Ronak Pradeep, Scott Miller,

Paul McNamee, and Luca Soldani. 2025. NeuCLIR-Bench: A modern evaluation collection for monolingual, cross-language, and multilingual information retrieval. *arXiv preprint arXiv:2511.14758*.

Dan Li and Evangelos Kanoulas. 2017. Active sampling for large-scale information retrieval evaluation. In *Proceedings of the 2017 ACM on Conference on Information and Knowledge Management*.

Nitika Mathur, Timothy Baldwin, and Trevor Cohn. 2020. Tangled up in BLEU: Reevaluating the evaluation of automatic machine translation evaluation metrics. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 4984–4997.

Harrie Oosterhuis, Rolf Jagerman, Zhen Qin, Xuanhui Wang, and Michael Bendersky. 2024. Reliable confidence intervals for information retrieval evaluation using generative A.I. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*.

Ricardo Rei, José G. C. de Souza, Duarte Alves, Chrysoula Zerva, Ana C. Farinha, Taisiya Glushkova, Alon Lavie, Luisa Coheur, and André F. T. Martins. 2022. COMET-22: Unbabel-IST 2022 submission for the metrics shared task. In *Proceedings of the Seventh Conference on Machine Translation (WMT)*, pages 578–585.

Tetsuya Sakai. 2016. Topic set size design. *Information Retrieval Journal*, 19(3):256–283.

Christoph Sawade, Niels Landwehr, and Tobias Scheffer. 2012. Active comparison of prediction models. In *Advances in Neural Information Processing Systems*.

Shivani Upadhyay, Ronak Pradeep, Nandan Thakur, Nick Craswell, and Jimmy Lin. 2024. UMBRELA: UMbrela is the (open-source reproduction of the) Bing RELevance Assessor. *arXiv preprint arXiv:2406.06519*.

Ellen Voorhees, Tasmeer Alam, Steven Bedrick, Dina Demner-Fushman, William R. Hersh, Kyle Lo, Kirk Roberts, Ian Soboroff, and Lucy Lu Wang. 2020. TREC-COVID: Constructing a pandemic information retrieval test collection. *ACM SIGIR Forum*, 54(1).

An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, et al. 2025. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*.

Yanzhao Zhang, Mingxin Li, Dingkun Long, Xin Zhang, Huan Lin, Baosong Yang, Pengjun Xie, An Yang, Dayiheng Liu, Junyang Lin, Fei Huang, and Jingren Zhou. 2025. Qwen3 embedding: Advancing text embedding and reranking through foundation models. *arXiv preprint arXiv:2506.05176*.

Zheng Zhang, Lufei Li, Xinyue Tan, Yuanhao Zeng, Ziwei Shan, Yexin Li, and Kan Ren. 2026. From judgment quality to downstream utility: Rethinking LLM-as-a-judge for open-ended tasks. *arXiv preprint arXiv:2609.37145*.

Tijana Zrnic and Emmanuel J. Candès. 2024a. Active statistical inference. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 62993–63010. PMLR.

Tijana Zrnic and Emmanuel J. Candès. 2024b. Cross-prediction-powered inference. *Proceedings of the National Academy of Sciences*, 121(15).

A Locked cells, three levels and retrieval

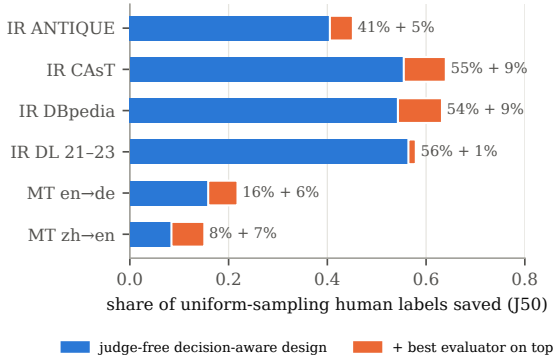


Figure 3: The four pre-registered MT cells and the four retrieval collections: share of uniform-sampling human labels (J_{50}) saved by the best fixed judge-free design and added by the best evaluator on the same design (retrieval: Qwen3-8B, $\varepsilon = 0.02$; MT: best of 31 metric variants post hoc, pilot-fixed coefficient, $\varepsilon = 0.01$).

Figure 4 places the 30 positively correlated WMT22 metric variants (REUSE omitted) on three levels against their realised HES. At the global level (Pearson over the 11 WMT submissions) nearly all strong metrics are above 0.9. Among the four closest systems the ranking is fragile: MetricX-XL-DA and MetricX-XXL-DA reach system-level Pearson 0.994–0.995 on zh→en and Kendall $\tau = 0$ among the top four. The second- and third-ranked systems differ by only 0.0009 (en→de) and 0.0017 (zh→en) in utility (standard error 0.002–0.004), so $\tau = 0$ can reflect fragility; but 10 of the 31 zh→en variants (system-level Pearson 0.83–0.96; BLEU, spBLEU, chrF, BERTScore, YiSi, SEScore, MEE) reach $\tau = -1$, which also reverses the first-ranked system's lead over each of the other three (2.3–3.5 standard errors). The quantity that governs a control variate is the correlation ρ of per-item paired differences, and it is low everywhere: 0.30–0.38 for the best metrics on en→de (mean over the menu's pairs), 0.21–0.25 on zh→en, and 0.17–0.29

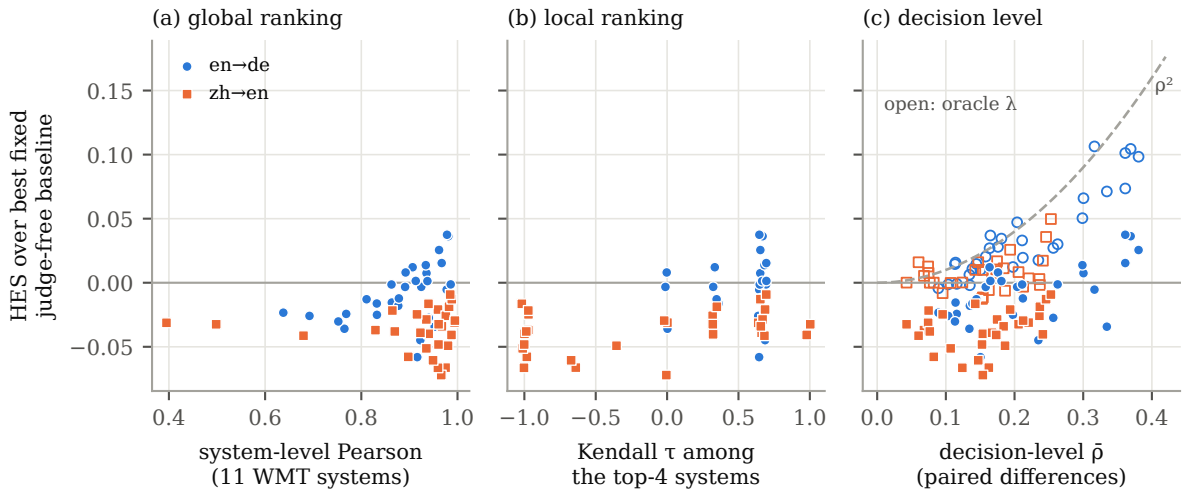


Figure 4: Thirty WMT22 metric variants at three levels of quality against their realised human-evidence saving over the best fixed judge-free baseline (pilot 50, mean over $\varepsilon \in \{0.01, 0.02\}$). (c) open markers: the same metrics with the population-optimal (oracle) correction coefficient; dashed: ρ^2, the asymptotic saving. REUSE (negative system-level correlation; switched off by $\lambda \geq 0$) is omitted.

Cell	uniform J_{50}	judge-free best fixed	+ evaluator best (median)	evaluator HES
IR ANTIQUE	1,405	836	770	8%
IR CAsT 2019	4,778	2,127	1,720	19%
IR DBpedia-Entity	5,563	2,543	2,046	20%
IR DL 21–23	4,092	1,783	1,722	3%
MT en→de, ε=.01	463	389 (w)	362 (390)	7% (0%)
MT en→de, ε=.02	261	238 (w)	233 (242)	2% (−2%)
MT zh→en, ε=.01	2,990	2,738 (d)	2,539 (2,760)	7% (−1%)
MT zh→en, ε=.02	1,166	1,066 (d)	1,052 (1,104)	1% (−4%)

Table 2: Unique human labels (J_{50}, pilot included) for uniform sampling, the best fixed judge-free design, and the *same* design with an evaluator added as control variate. Retrieval: Qwen3-8B judge, $\varepsilon = 0.02$, pilot 20 queries. MT: the best of the 31 WMT22 metric variants chosen post hoc, and the median metric in parentheses, on the cell's best fixed design (w: weighted, d: dedup); pilot 50 segments, pilot-fixed coefficient. 95% Monte-Carlo intervals: retrieval HES [5, 10], [14, 24], [13, 27] and [1, 7]%; MT design savings [10, 29], [−13, 15], [3, 14] and [6, 15]%, best-metric HES [−7, 12], [−15, 6], [−6, 19] and [−3, 8]%: single MT cells are weak evidence, and Figure 2 pools 33. H8 as locked used the weighted design for every metric (App. E).

for COMET-22 and the Qwen3-8B GEMBA judge (system-level 0.97–0.99; here medians over pairs of the top-6 systems), while the Mistral-7B GEMBA judge reaches only 0.04–0.09. Part of this is label noise: in the en→de three-rater re-annotation, two raters' paired differences on the same items correlate only 0.31–0.41, so even a perfect evaluator would reach $\rho \approx 0.59$ against the single ratings an audit collects; the best metrics attain one half to two thirds of that (App. E). Across metrics, HES is ranked slightly better by ρ than by system-level Pearson (Spearman 0.39 vs. 0.36 on en→de, 0.22 vs. 0.07 on zh→en; H11), but neither predicts it well: all real metrics sit where the saving is small (§4).

Retrieval setup. The retrieval records come from earlier, unpublished experiments and are re-analysed here without new labelling or inference. Per collection, a shared candidate pool is the union of the top 30 documents of BM25 and three sentence-transformer retrievers (all-MiniLM-L6-v2, all-mpnet-base-v2, msmarco-MiniLM-L6-cos-v5), and the menu holds three cutoff policies on it: an expected-F1 gate with a target-calibrated scalar, a global score threshold, and a learned truncation (the gate's relevance model is a calibrated logistic regression on four pool features, trained on four BEIR development collections and never on a target). A policy's utility on a query is its precision at its cutoff under the collection's human relevance judgments. An audit labels a 20-query pilot in full, takes the pilot's best policy as the candidate, labels a pre-committed number of further documents and certifies the candidate if a one-sided t bound ($\alpha = 0.10$, Bonferroni-corrected) on its regret to every alternative is below ε; documents outside every cutoff carry no decision weight and are never labelled. The judge-free design samples documents in proportion to their closed-form decision weight, and the evaluator enters as a control

variate on that design with $\lambda \in [0, 1]$ fitted on the pilot. J_{50} counts unique human labels, pilot included, as in §2; each record is one simulated audit (300 per cell), and J_{50}, HES and their paired-bootstrap intervals are recomputed from the per-draw records by `code/101_ir_hes.py` (outputs in `results/ir`). The 54 settings below are (collection, policy pair, judge) combinations over four BEIR development pools under two retrieval stacks and five fully judged pools, with Qwen3-Reranker-0.6B and Qwen3-8B as judges; there the gain is the ratio of the squared estimated standard errors of the human-only and control-variate estimators of a paired policy difference over 300 random audits of 10 queries with a cross-fitted coefficient.

In the retrospective retrieval analysis, the evaluator's gain across 54 experimental settings correlated 0.75 with ρ and 0.03 with label accuracy; on the three collections held out in those experiments (NeuCLIRBench, TREC-COVID, Touché-2020; Lawrie et al., 2025; Voorhees et al., 2020; Bondarenko et al., 2020) the design saves 6–50% of uniform-sampling labels and the judge adds HES 2–16%; in the locked NeuCLIRBench test a 20-query pilot over-estimated the judge's increment (cost ratio 0.71–0.76 predicted, 0.82–0.93 realised).

B Derivation of the cost approximation

Write D for the paired difference of an informative item, $\sigma^2 = \mathrm{Var}(D)$, and $\hat{D}$ for the evaluator's prediction with $\mathrm{Corr}(D, \hat{D}) = \rho$. For a fixed coefficient λ the residual $R = D - \lambda\hat{D}$ has variance $\sigma^2 - 2\lambda\rho\sigma\hat{\sigma} + \lambda^2\hat{\sigma}^2$, minimised at $\lambda^\star = \rho\sigma/\hat{\sigma}$ to $\sigma^2(1-\rho^2)$ (Angelopoulos et al., 2023b). With $\hat{\lambda}$ the least-squares coefficient from a pilot of effective informative size P_{eff}, independent of the post-pilot sample, and a homoskedastic residual ($\mathrm{E}[R^2 \mid \hat{D}] = \sigma^2(1-\rho^2)$), $\mathrm{E}[(\hat{\lambda} - \lambda^\star)^2] \approx \sigma^2(1-\rho^2)/(P_{\text{eff}}\hat{\sigma}^2)$, so the expected residual variance is $\sigma^2(1-\rho^2)(1 + 1/P_{\text{eff}})$ (Cochran, 1977, §7). Under the normal fixed-budget approximation the post-pilot labels needed to bring $z\sqrt{v/L}$ below the slack scale with the variance factor, so $L \approx L_0(1-\rho^2)(1 + 1/P_{\text{eff}})$ and $1 - L/L_0 \approx \rho^2 - (1-\rho^2)/P_{\text{eff}}$ to first order in $1/P_{\text{eff}}$. With the pilot cost B_{pilot} common to both arms, $\text{HES} = 1 - (B_{\text{pilot}} + L)/(B_{\text{pilot}} + L_0) = (1 - L/L_0)(1 - B_{\text{pilot}}/J_0)$. Clipping λ at zero bounds the loss of an evaluator with $\rho \le 0$ by the variance of $\hat{\lambda}$ only; zero-difference items reduce P_{eff} below n_{pilot}, which is why MT, with about half its paired differences zero, pays more tax. In general, $\mathrm{Var}(\hat{\lambda}) \approx \mathrm{E}[(\hat{D} - \mathrm{E}\hat{D})^2 R^2]/(P_{\text{eff}}\,\mathrm{Var}(\hat{D})^2)$ (sandwich form), which reduces to the expression above under homoskedasticity; heavy-tailed or heteroskedastic residuals, as with sparse MQM differences, raise the tax. The coefficient that minimises the variance of (1) under a given design, $\sum_i (1-\pi_i)/\pi_i\,(D_i - \lambda\hat{D}_i)^2$, is a least-squares slope through the origin with weights $(1-\pi_i)/\pi_i$. Under uniform sampling the weights are constant and it is $\sum_i D_i\hat{D}_i / \sum_i \hat{D}_i^2$, which coincides with the covariance-over-variance coefficient λ_j of (1) only when $\hat{D}$ has mean zero over the items (or an intercept is fitted); in general the two differ by a term in the means, and the paper's λ_j is the covariance-over-variance form throughout. Empirically the difference is small: on the weighted design of the top-4 menus (31 metrics, three semi-synthetic evaluators, 20 pilots) the variance with λ_j exceeds the design-optimal minimum by a median 0.0%, a 90th percentile of 0.2% and at most 3.7%, so ρ^2 is used as an interpretive approximation, not a bound.

C The estimation tax on real evaluators

The estimation tax on real metrics. Replacing the pilot-fitted coefficient by the population-optimal one isolates the tax. Over the 31 metric variants it is 4.3 [3.5, 5.2] points on en→de and 2.5 [2.0, 3.1] on zh→en (metric-level bootstrap; H9). For the five metrics with the largest ρ^2 (0.11–0.15 on en→de, as $(\bar{\rho})^2$ or as PPSR's mean ρ^2, within the 0.09–0.15 PPSR reports), the realised saving over uniform sampling is between -4% and $+6\%$ (en→de) and -6% and $+1\%$ (zh→en), below the correlation-based saving in every cell (H10, under both definitions). With the oracle coefficient the same en→de metrics reach 6–12% on the decision-aware design: the asymptotic saving is not wrong, it is what an auditor who already knew λ would obtain.

Is the tax an artefact of freezing λ? Mostly. Standard PPI++ fits λ on all labelled items. We reran the eight strongest metric variants per language pair, COMET-22 and the two GEMBA judges with the coefficient refitted on the pilot and every post-pilot label, and cross-fitted over two folds (App. D). Mean HES on the weighted design rises from $+0.4\%$ (pilot coefficient) to $+6.8\%$ with refitting and $+1.6\%$ with cross-fitting on en→de, against $+6.7\%$ for the oracle coefficient; on zh→en from -0.4% to $+2.0\%$ and $+1.1\%$, against $+1.3\%$;

on the six chat pairs from $+1.1\%$ to $+2.3\%$ and $+1.5\%$, against $+2.1\%$. Its variance estimate ignores the coefficient's estimation error, but over the 30 decisions and the six chat pairs the refitted upper bound covered the true difference at 0.90 (nominal 0.90), also at the budget nearest each design's J_{50}, where it certifies (refit 0.902, cross-fit 0.903, pilot 0.903, human-only 0.904), and under boundary stress its largest type-I error was 0.05. Cross-fitting recovers less because each fold's coefficient rests on half of the labels; a practitioner who needs no budget plan can refit, one who wants the variance estimate to account for the coefficient should cross-fit or freeze (the bound itself is calibrated by simulation, not proved). Refitting can exceed the oracle, which is the population covariance-over-variance coefficient (App. B); on the weighted design the coefficient that minimises the realised Poisson-design variance differs from it by a median 0.0% and at most 3.7% of that variance (App. B), so the two are interchangeable here. Across the 30 decisions (Figure 2b) refitted HES follows the tax-free approximation $\rho^2(1 - B_{\text{pilot}}/J)$ (median deviation -0.4 points, correlation 0.42 over 1,158 decision–evaluator points), whereas with the pilot coefficient it falls a median 2.3 points below it; 94% of the approximate values are below 5%, because decision-level ρ is at most 0.43 and the pilot is a median 39% of the cost (range 15–66%). The tax is therefore the price of committing to the coefficient before the audit (to plan the budget, say), not a property of the evaluator; refitting remains exploratory (the same labels set the coefficient and the inference). Even refitted, in the four locked cells the mean of the eleven (6.9, 6.7, 4.8, -0.3%) stays below the design (15.9, 9.0, 8.4, 8.5%).

Pilot size trades tax for fixed cost. Equation (2) predicts that a larger pilot lowers the tax $(1 - \rho^2)/P_{\text{eff}}$ but raises the fixed cost B_{pilot}. Both effects appear. Doubling the MT pilot to 100 segments shrinks the losses of the weak LLM judges (Mistral-7B, en→de, $\varepsilon = 0.01$: $-7.1\% \to -0.9\%$) and halves the saving of a semi-synthetic $\rho = 0.9$ evaluator (32%→15%) because the pilot becomes most of the cost. A smaller pilot does not rescue the evaluators either: with pilots of 25 and 10 segments on the 30 two-system decisions (49 and 60 informative cells, now including en→de; pilot share 27% and 13% of the cost; the 31 metric variants), the design saves a median 4.3% and 4.4%, the mean evaluator adds 0.7% and 0.3% refitted (-2.7% and -7.7% with the pilot coefficient, whose tax grows as $1/P_{\text{eff}}$), and the design saves more in 86% and 90% of cells (App. E).

The finite-pilot statement behind (2):

Proposition 1 (Finite-pilot cost approximation)**.** *Let a design need L_0 post-pilot labels for a fixed decision without an evaluator, and let the evaluator's paired differences $\hat{D}$ have correlation ρ with the human ones D. Assume that, over informative items (those with a non-zero difference), D is a linear projection on $\hat{D}$ with a residual whose variance does not depend on $\hat{D}$ (homoskedastic; joint normality is a sufficient case), and that λ is fitted by least squares on a pilot independent of the post-pilot sample, with effective informative size P_{eff} (the number of pilot items that contribute variation to the paired regression, approximated by those with a non-zero paired difference; λ_j and P_{eff} are per competitor j). Then, to first order in $1/P_{\text{eff}}$ and under the normal fixed-budget approximation, the variance of the corrected estimator is $(1 - \rho^2)(1 + 1/P_{\text{eff}})$ times that of the uncorrected one and*

$$\text{HES} \approx \left[\rho^2 - \frac{1 - \rho^2}{P_{\text{eff}}}\right]\left(1 - \frac{B_{\text{pilot}}}{J_0}\right), \quad (3)$$

with $J_0 = B_{\text{pilot}} + L_0$.

D Robustness of the estimation tax

Mean HES over the evaluators of each set, over their own judge-free base, by how the coefficient is set (J_{50}; MT: eight strongest metric variants per language pair, COMET-22, Qwen3-8B, Mistral-7B; chat: Qwen3-8B, Mistral-7B, longer). Pilot: fitted on the pilot and frozen (the frozen-coefficient baseline); refit: refitted by unweighted least squares on the pilot and every post-pilot label; xfit: cross-fitted over two folds; HT: the same with $1/\pi$-weighted least squares; oracle: population-optimal. COMET-22 enters twice in MT, as the WMT22 submission (COMET-22-refA) and as our own run; their segment scores correlate only 0.78–0.79, and dropping ours changes the MT means by at most 0.3 points.

Cell	base	pilot	refit	xfit	refit_{HT}	xfit_{HT}	oracle
MT en→de	weighted	0.4	6.8	1.6	10.0	−1.3	6.7
MT en→de	dedup	0.8	7.2	1.7	12.4	−1.3	–
MT zh→en	weighted	−0.4	2.0	1.1	2.0	1.1	1.3
MT zh→en	dedup	−1.8	2.3	1.9	2.5	2.3	–
Chat (6 pairs)	uniform	1.1	2.3	1.5	2.3	0.9	2.1

Values in %, averaged over ε. With J_{80} in place of J_{50} the MT values are 0.1–1.2 (pilot), 2.2–4.9 (refit), 1.5–2.8 (xfit), 2.1–4.4 (refit_{HT}), 0.7–2.9

(xfit$_{\mathrm{HT}}$) and 1.9–2.5 (oracle); J_{80} is often near the largest budget, so these are less reliable. Under boundary stress the largest type-I errors on en→de were 0.04 (pilot), 0.047 (refit), 0.043 (xfit), 0.07 (refit$_{\mathrm{HT}}$), 0.05 (xfit$_{\mathrm{HT}}$) and 0.06 (human-only); no cell was significantly above α. Evaluator-driven allocation (sampling by the pilot-estimated residual variance, as in active inference) reduced J_{50} relative to the weighted design with the same evaluator by -1.6% to $+1.7\%$ on average over the 31 metric variants (individual cells -11% to $+19\%$).

E Sensitivity of the decision analysis and a human noise ceiling

All exploratory and post-lock (`code/123_decision_sensitivity.py`, `code/124_human_ceiling.py`).

Informativeness threshold. The post-pilot share $1 - B_{\mathrm{pilot}}/J$ of the 60 MT cells takes no value between 0.21 and 0.47, so every threshold in that range selects the same 33 cells. With a threshold of 0.2 (34 cells), 0.5 (30) or 0.6 (23; one en→de cell) the design's median saving is 5.4, 5.8 and 5.2%, the mean evaluator's refitted HES 1.5, 1.5 and 1.7%, and the design beats the mean evaluator in 94/82/91, 93/83/90 and 91/78/87% of cells (pilot/refit/cross-fit).

Evaluator set. Restricting the mean to the ten evaluators per language pair with the highest system-level Pearson, chosen without reference to any decision, raises the mean evaluator's median HES on the 33 cells to 0.6, 2.8 and 1.8% (pilot, refit, cross-fit); the design still saves more in 91, 76 and 79% of cells.

One rating per identical string. WMT22 rated identical strings separately, and our estimand averages their ratings. To check that the deduplication saving does not depend on this averaging, each group of identical strings instead shares one of its ratings, drawn at random (`-ident pick`). In the four locked cells the best fixed design then saves 17.0, 8.5, 9.2 and 12.2% (means over three draws of the shared rating; ranges 15.5–18.4, 7.9–8.8, 7.9–10.0 and 11.0–13.9) instead of 15.9, 9.0, 8.4 and 8.5%. Over the 30 decisions (again 33 informative cells) the design's median saving is 5.2% instead of 5.5%, the mean evaluator's median HES -0.6, 1.4 and 0.1% (pilot, refit, cross-fit) instead of -0.1, 1.6 and 0.4%, and the design beats it in 91, 85 and 91% of the cells instead of 94, 82 and 91%. The deduplication saving therefore does not rest on the averaging.

Dissimilarity bins. The weighted design bins the positive chrF dissimilarities into three groups (below the median, third and fourth quartile; identical outputs have weight zero). With four quartile bins instead, the design's J_{50} in the four locked cells changes by at most 0.8% (weighted: 390 vs. 389, 238 vs. 238, 2,743 vs. 2,767 and 1,103 vs. 1,095 labels), the best fixed design's saving by at most 0.1 points, and the best metric's increment by at most 1.2 points; over the 30 two-system decisions (the same 33 informative cells, 31 metric variants) the design's median saving is 5.5% under both binnings and the mean evaluator's HES -0.5 / 1.3 / 0.2% (pilot / refit / cross-fit) against -0.1 / 1.6 / 0.3% with three bins (supplementary `results/VARIANTS.md`).

Monte-Carlo precision. With 2,000 audits per cell (31 metric variants) the informative cells are 36 rather than 33 (en→de decisions 13 and 24 at $\varepsilon = 0.01$ and zh→en 05 at $\varepsilon = 0.02$ cross the 0.3 threshold, with post-pilot shares 0.37–0.42); the design's median saving is 4.6% (300 audits: 5.5%), the mean evaluator's 0.2 / 2.3 / 0.9% (-0.1 / 1.6 / 0.4%), and for the ten strongest evaluators 1.4 / 3.9 / 2.2% (0.6 / 2.8 / 1.8%); the design saves more in 100 / 89 / 100% of cells (89 / 61 / 78% against the ten strongest). The one-sided bounds of §3.2 are in supplementary `results/EXCLUSION_n2000.md`. The 2,000-audit runs give the WMT22 numbers of the abstract and §3.2; the 300-audit runs, which also include the three LLM judges (34 evaluators), give Figure 2 and the WMT22 panel of Figure 1. Their differences are those listed above.

Pilot size. The 30 two-system decisions re-run with pilots of 25 and 10 segments (the 31 metric variants; the same menus, ε and coefficient rules). Informative cells: 49 and 60 (pilot 50: 33); pilot share of the cost, median: 27% and 13% (35%); design saving, median: 4.3% and 4.4% (5.5%); mean evaluator HES, median over cells, pilot / refit / cross-fit: -2.7 / 0.7 / 0.0% and -7.7 / 0.3 / 0.0% (pilot 50: -0.1 / 1.6 / 0.4%); for the ten evaluators with the highest system-level correlation -2.4 / 2.2 / 0.0% and -6.8 / 0.5 / 0.0%; the design saves more than the mean evaluator in 98 / 86 / 92% and 100 / 90 / 98% of cells. The frozen-coefficient tax grows as the pilot shrinks, as (2) predicts, and refitting leaves the increment below 1%.

Evaluators on the weighted vs. the dedup base. Lock v1.1 applied every metric on the weighted design and compared it with the cheapest judge-free design, which on zh→en is dedup; Table 2 now applies the metrics on that same base. On zh→en at $\varepsilon = 0.01$ the best metric adds 7.3% [−6, 19] on dedup against -0.9% on weighted; at $\varepsilon = 0.02$, 1.3% against 0.8%. The H8 verdict is unchanged either way, and the locked numbers are kept in `results/MTME_VERDICTS.md`.

Human noise ceiling. WMT22 re-annotated en→de with three raters per item. For each of the 15 pairs of the top six systems, the paired differences of two distinct raters (drawn at random per item, 200 repetitions) correlate 0.31–0.41 (median 0.35; the same without identical outputs). Since the audits' labels are single ratings, an evaluator that knew the true difference would reach at most $\rho \approx \sqrt{0.35} = 0.59$ (0.55–0.64 per pair) against them, and $\rho^2 \approx 0.35$ caps its tax-free saving at about a third of the post-pilot labels. The best en→de metrics (ρ 0.30–0.38) reach one half to two thirds of that ceiling, and their ρ^2 a quarter to two fifths of it, so both label noise and the evaluators limit the saving.

F Pre-registration record

Pre-registration. Lock v0.8 (MT, chat) was committed before any LLM or COMET judgment was joined with a human label; v0.9 replaced the uninformative v0.8 chat decision (gaps 0.10–0.56, settled by the pilot alone) by close pairs before they were judged; v1.1 fixed the 31-metric test before any of its savings were computed. All locks were committed within about four hours on public data whose human labels exist; before v0.8 the human-only MT designs had been run in development (they saved 8–19% over uniform sampling and chrF about nothing, so H1 was not a blind test), and v1.1 followed the exploratory estimation-tax analyses of the open judges, so H9 tests on 31 new evaluators a phenomenon first seen on four. All locks, deviations and outcomes are in the supplementary material.

Table 3 lists every pre-registered hypothesis. The design-dominance and estimation-tax hypotheses held; H2, H5 and H6 did not, and H4 held only in MT. H2 (value over uniform sampling inflated by ≥5 points) failed because the open judges' value was near zero on both baselines; H5 (selection by pilot-predicted cost beats selection by pilot accuracy) did worse in both domains (excess cost 3.2% vs. 2.3% in MT, 2.8% vs. 1.7% in chat; near break-even, a 50-item pilot cannot resolve a few points). H4 (pilot predicts relative design cost) held in MT (0.82), not in chat (0.37).

	Hypothesis	Outcome
H1	design saving > best open judge (MT)	4/4
H2	inflation ≥ 0.05 (MT)	**0/4**
H3	HES tracks ρ > accuracy[a]	held[d]
H4	pilot predicts cost ($r \geq .8$)	MT; **chat no**
H5	pilot selector beats accuracy	**no (both)**
H6	chat HES ≥5% in ≥3/6	**2/6**
H8	design ≥ best of 31 variants[b]	4/4
H9	estimation tax > 0	both pairs
H10	realised HES < ρ^2 (top 5)[c]	all cells
H11	HES tracks ρ > system r	both

Table 3: Pre-registered hypotheses (locks v0.8, v0.9, v1.1). [a]Spearman over two-system menus. [b]with the locked pilot-fixed coefficient; with a refitted coefficient the post-hoc best of the eleven strong evaluators exceeds the design in 2/4 cells. [c]largely implied by the dilution factor $1 - B_{\text{pilot}}/J$. [d]by the locked rule; in MT both correlations are near zero (0.07 vs. -0.03), so we read it as inconclusive there. H7 and H2′ concern frontier judges and are recorded in Appendix F.

Validity. Over all locked cells the observed wrong-certificate rate is at most 0.070 (Clopper–Pearson upper limit 0.105). Under boundary stress (the runner-up as candidate, ε at 0.9 times its regret) the largest observed type-I error among 882 chat cells is 0.143 at nominal 0.10 (0.14 on the v0.9 set); no cell is significantly above α after Bonferroni correction (0.5% of cells at $p < 0.05$), and a Student-t bound changes the maximum to 0.140. These are Monte-Carlo checks of an asymptotic bound, not finite-sample guarantees.

Lock v0.8 (MT, chat; hypotheses H1–H5 and validity) was committed before any LLM or COMET judgment was joined with a human label; the human-only MT designs and chrF had been examined in development and are reported as such. Two deviations followed (a type fix and an accounting fix charging one Arena vote per battle). Lock v0.9 replaced the v0.8 chat decision, whose pairs had win-rate gaps of 0.10–0.56 and were certified by every design right after the pilot, by the six closest pairs; no judge output existed for these battles. Lock v1.1 fixed the 31-metric test (H8–H11) before any of its savings were computed; H10 as locked uses $(\bar{\rho})^2$ and is reported with PPSR's mean of squared correlations as a post-lock robustness check, and H9's interval is a metric-level bootstrap.

Lock v1.0 (frontier judges) is committed with its exported inputs and scoring rule and has not been run. Analyses not listed in a lock (three-level anatomy, oracle-λ tax for open judges, presentation order, semi-synthetic ρ) are exploratory.

G Controlled variation of ρ

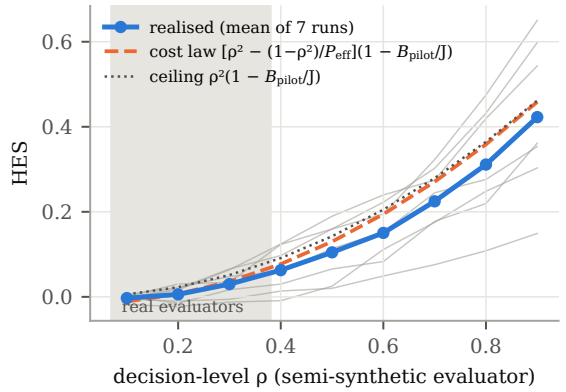


Figure 5: Realised HES of semi-synthetic evaluators with a set decision-level ρ (grey: individual runs) against the cost law (2) and its oracle-λ value. Shaded: the range of the real evaluators. Identical output strings of an item share one noise draw (an earlier version drew noise per output, which left the never-sampled known-zero items with a non-zero prediction term of at most 6×10^{-4}; the correlation was 0.975 and the savings 11% / 32%).

H Held-out year: auditor strategies on WMT23

Pre-registered (`prereg/PROSPECTIVE_LOCK_v1.2.md`) after the WMT22 development analysis and before any WMT23 record was examined; `code/128_strategies.py`, `results/HELDOUT_WMT23.md`. WMT23 pools use the official mt-metrics-eval MQM scores (equal to the sum of Google's merged error weights in 100% / 98.5% of items; one rater per item); references and the BLEU-MBR system are excluded as in WMT22. Every strategy is a per-audit choice of one arm from the same simulated audits, made with pilot information only; cost is J_{50} with the pilot included. On WMT22 the dedup design was cheaper than the weighted one in 83% (zh→en) and 75% (en→de) of informative cells, so S1 is dedup; COMET-22 had the highest decision-level ρ among metrics submitted in both years (MetricX-22 has no WMT23 counterpart); pilot ρ and the pilot-predicted label count gave the same development result as selection statistics; a predicted-saving threshold only removed gains and is not used. Table 1 in the main text summarises the outcome.

Direct comparison of S2 and S3 (addendum to lock v1.2). The 300-audit results could not say whether pilot selection differs from the pre-fixed evaluator. We therefore declared, before running, a paired comparison $G_c = 1 - J_{50}(\mathrm{S3}, c)/J_{50}(\mathrm{S2}, c)$ on the same audits with a paired bootstrap over audits, a cluster bootstrap over decisions for the pooled mean (now the sensitivity analysis below), a margin of 2 points of J_{50} (the Monte-Carlo half-width expected at 2,000 audits and the size of the smallest savings the paper calls real), and a re-run of the 30 WMT23 decisions at 2,000 audits and in boundary stress (`code/129_s2_vs_s3.py`, `results/strategies/S2_vs_S3_*`). The selection-noise test of §3.2 compares the post-hoc best with the mean evaluator; it is not a test against S2, and the post-hoc best is a reference, not a strategy. Table 4: means over cells in points of J_{50} (positive: the first strategy is cheaper), refitted coefficient. The primary interval (MC) is Monte-Carlo only: the benchmark is fixed, one resampling of audit indices is applied to every cell at once, because the audits of different cells share the simulator's random streams (same seed and draw index), and each replicate recomputes exactly the reported statistic, the mean over cells. The cluster interval treats the decisions as clusters (both ε of a decision together) and combines a cluster bootstrap with the joint draws; its point estimate is the mean over decisions of the within-decision mean, which differs from the cell mean where a decision keeps one or two informative cells (-2.1 and $+5.2$ in place of -1.5 and $+4.4$ on WMT22; within 0.1 point on WMT23). An earlier version paired a cell-mean estimate with a cluster interval. WMT22: 300 audits per cell and 400 resamplings, from the re-run with known-zero differences set to zero (tag `pairmtmez`; every WMT22 entry of Table 1 agrees with the original run within 0.1 point). WMT23: 2,000 audits and 1,000 resamplings (tag `pair23n`; the 300-audit run gave $G = +0.2$ with a cluster interval $[-2.7, +3.3]$, undecided at the margin). Per-cell intervals have a median width of 15–18 points on WMT22 and 4–6 on WMT23, where two of the 26 cells have a G interval above zero and none below, and S2 beats S1 in 21. The 2-point margin was set after

cells	year	n	G (S3 vs S2)	MC 95%	clusters 95%	S2 vs S1	S3 vs S1
informative	WMT22	33	-1.5	$[-4.4, +0.3]$	$[-8.9, +1.3]$	$+4.4$ $[2.1, 6.7]$	$+3.2$ $[0.7, 5.1]$
informative	WMT23	26	-0.1	$[-0.7, +0.5]$	$[-1.2, +1.1]$	$+3.4$ $[2.9, 3.9]$	$+3.3$ $[2.7, 3.9]$
all 60	WMT22	60	-2.1	$[-3.6, -0.5]$	$[-4.5, 0.0]$	$+2.4$ $[1.2, 3.8]$	$+0.5$ $[-0.6, 2.3]$
all 60	WMT23	60	-0.1	$[-0.3, +0.2]$	$[-0.6, +0.5]$	$+1.8$ $[1.5, 2.0]$	$+1.7$ $[1.4, 2.0]$

Table 4: S2 against S3 and both against S1 on the same audits; means over cells in points of J_{50} (positive: the first strategy is cheaper), refitted coefficient, unrestricted protocol. MC: Monte-Carlo interval from the joint resampling of audit indices; clusters: decisions as clusters.

the 300-audit results and before the 2,000-audit run; it is a one-sided check that selection does not gain 2 points, not a pre-registered equivalence test.

Coverage and errors of the strategies. Coverage of the chosen arm's one-sided bound (nominal 0.90) over the informative cells, mean over cells (WMT22 / WMT23): at the budget nearest J_{50}, S1 0.90 / 0.90, S2 0.90 / 0.90, S3 0.89 / 0.89; averaged over all budgets, S1 0.90 / 0.91, S2 0.90 / 0.91, S3 0.90 / 0.90; at the worst budget of a cell, S1 0.86 / 0.89, S2 0.86 / 0.88, S3 0.86 / 0.88 (minima over cells 0.83 / 0.86). Wrong-certificate rates at J_{50} are 0.003 (WMT22) and 0.006 (WMT23) for every strategy, maxima over budgets 0.007–0.008 and 0.009–0.012. These are the records with known-zero differences set to zero. On the records without it (the 300-audit runs of both years reported in an earlier version of this paper) the refitted strategies covered at 0.86–0.88 averaged over all budgets and S2 at 0.45–0.50 at the worst budget of a cell (minimum 0.15): where two outputs coincide the human difference is exactly zero while 13–14 WMT22 and 26–27 WMT23 metrics report a small non-zero one, so the term $\lambda \sum_{\text{rest}} \hat{D}_j$ of (1) carries a bias over the unsampled known-zero items that a coefficient fitted on few labels amplifies. Zeroing them restores nominal coverage at the same costs: on one WMT23 cell re-run with the same seed, J_{50} is unchanged to the first decimal for S1 and S2 and within one label for S3 while S2's coverage at the smallest budget goes from 0.43 to 0.86 (`results/strategies/KNOWN_ZERO_COVERAGE_CHECK.md`), and on the WMT22 re-run every strategy's excess over S1 is within 0.1 point of the original (`S2_vs_S3_wmt22.md` against `S2_vs_S3_wmt22z.md`).

Boundary stress and the budget range of nominal error. The 30 WMT23 decisions re-run with the runner-up as candidate and ε at 0.9 times its true regret (300 audits per cell, tag `pair23b`, `S2_vs_S3_wmt23_boundary.md`): every certificate is wrong, so the certification rate is the type-I error at nominal 0.10. Budgets are expected numbers of sampled items under the Poisson design; a post-pilot budget of 10 items is about 116 (en→de) and 110 (zh→en) labels in total, pilot included. The rate of a budget is the mean over the 15 cells of a language pair, with a 95% Monte-Carlo interval from the joint resampling of audit indices (the cells share random streams, so their audits are not pooled as independent trials). At 10 items: S1 0.18 [0.16, 0.20] and 0.18 [0.17, 0.20], S2 0.16 and 0.19, S3 and S4 0.12 and 0.13 (en→de and zh→en). The observed mean rate is at or below 0.10 at every later budget from 55 items (206 labels; S1 0.092 [0.078, 0.105]) on en→de and from 88 items (265 labels; 0.087 [0.076, 0.099]) on zh→en, for S1 and S2 alike (S3: 55 and 42); it stays at or below 0.125 from 23 items (144 labels) and 29 items (148 labels). These are observations on this simulation, not tests that the error is below nominal: at the cut-offs the intervals still reach 0.10. It is 0.05 at about 450 labels and 0.01–0.02 at the largest budgets. The per-cell maximum over budgets averages 0.18 (S1), 0.18 (S2) and 0.13 (S3), at most 0.24. Against S1 on the same cell and budget, the evaluator strategies are lower on average (S2 by 0.7 and 0.1 points, S3 by 1.4 and 1.5 on en→de and zh→en), higher at 19–34% (S2) and 15–19% (S3) of the 210 cell–budget points of a pair, higher by more than 2 points at 1–3%, and at most 3.7 points higher. The excess at the smallest budgets is the normal bound on few items, as in App. F; it applies to the human-only baseline as much as to the evaluators. With the cut-offs fixed, a second boundary run of the 30 decisions with a different seed (tag `pair23c`, `S2_vs_S3_wmt23_boundary_seed2.md`) gives, at 55 and 88 items, S1 0.088 [0.075, 0.100] and 0.079 [0.067, 0.092], S2 0.083 [0.070, 0.095] and 0.084 [0.072, 0.097], and at 23 and 29 items S1 0.119 [0.103, 0.133] and 0.119 [0.103, 0.135]; at ten items 0.18 again. The cut-offs are therefore reproducible as observations, with intervals that still

touch the nominal level; they are not a calibrated bound.

Restricted protocol (post-hoc sensitivity analysis). The same minimum budget applied to every arm of the 2,000-audit WMT23 records (`-min_budget`): J_{50} is interpolated over the allowed budgets only, so it is at least the cost of the first allowed budget (an earlier version interpolated across the disallowed budgets). Cut-offs from the boundary run above: 55 / 88 items (en→de / zh→en; observed mean error at or below 0.10 at every later budget) and 23 / 29 items (at or below 0.125). The cell sets are fixed to the 26 informative cells of the unrestricted run and to all 60 (`-cells_from`), so the three protocols are compared on the same cells (Table 5). Under the 0.10 cut-off the first allowed budget costs 204–207 (en→de) and 240–272 (zh→en) labels, 22 of the 26 cells certify there under every strategy, and the saving survives only where J_{50} was already above it (en→de 01 at both ε and 34 at $\varepsilon = 0.02$: S2 3.0, 3.2, 3.3%). Under the 0.125 cut-off six cells tie and S2 is cheaper in 20. The 0.10 and 0.125 conditions are different conditions and are not pooled. WMT22 has no boundary run of the two-system decisions (its locked-cell boundary runs are in App. F), so the cut-offs are read off WMT23 only.

Same-condition decomposition. Medians over informative cells of the shares of uniform-sampling labels (J_{50}) saved, refitted coefficient (`results/strategies/DECOMPOSITION_n2000.md`; WMT22 from the zeroed 300-audit re-run, WMT23 at 2,000 audits): design s_D 5.2% (WMT22) and 3.8% (WMT23); evaluator on uniform sampling s_B and on dedup h: mean evaluator 2.1 / 1.8 and 2.9 / 2.5, COMET-22 4.6 / 2.9 and 3.3 / 3.1, strongest 5.0 / 3.7 and 5.5 / 4.7, post-hoc best 4.9 / 6.8 and 6.9 / 6.2; totals s_{total} 7.1, 9.0, 9.6, 12.2 (WMT22) and 7.3, 7.3, 9.3, 11.2 (WMT23). With a frozen coefficient the mean evaluator's h is 0.2 and 0.3, COMET-22's 1.4 and 0.8, the strongest's 1.2 and 2.3. The 300-audit WMT23 records without zeroing gave the same picture (design 4.1%, mean evaluator 2.4%, strongest 5.6%, post-hoc best 7.0%; `DECOMPOSITION.md`).

I Chat audits

Figure 6 shows the six close LMArena pairs of lock v0.9 (§3.1). On the six close Arena pairs the pairwise judges reach (pilot-median) decision-level ρ of -0.01 to 0.29, the length heuristic 0.07–0.21. The best judge saves at least 5% with an interval above zero in two pairs (Figure 6a in App. I; H6 required three). Position bias explains part of the low ρ: one presentation order gives ρ of 0.19 (Qwen3-8B) and 0.12 (Mistral-7B), both orders averaged 0.25 and 0.16, turning Mistral's saving from -1 to -2% into $+0.5\%$. The bias itself is corrected by the human labels; the noise it adds lowers ρ.

protocol	26 informative cells S2 vs S1	26 informative cells S3 vs S1	26 informative cells G	all 60 cells S2 vs S1	all 60 cells G
unrestricted	$+3.4\ [2.9, 3.9]$	$+3.3\ [2.7, 3.9]$	$-0.1\ [-0.7, +0.5]$	$+1.8\ [1.5, 2.0]$	$-0.1\ [-0.3, +0.2]$
cut-off at 0.10 (55 / 88 items)	$+0.4\ [0.2, 0.6]$	$+0.2\ [-0.1, 0.5]$	$-0.2\ [-0.4, +0.1]$	$+0.2\ [0.1, 0.2]$	$-0.1\ [-0.2, +0.0]$
cut-off at 0.125 (23 / 29 items)	$+3.0\ [2.5, 3.4]$	$+3.0\ [2.4, 3.5]$	$-0.0\ [-0.5, +0.5]$	$+1.3\ [1.1, 1.5]$	$-0.0\ [-0.2, +0.2]$

Table 5: Post-hoc sensitivity analysis: the strategies under a protocol that issues no certificate below a minimum post-pilot budget (en→de / zh→en items), the same rule for every strategy, on fixed cell sets. Means in points of J_{50} with Monte-Carlo intervals (`S2_vs_S3_wmt23_n2000_restricted*.md`).

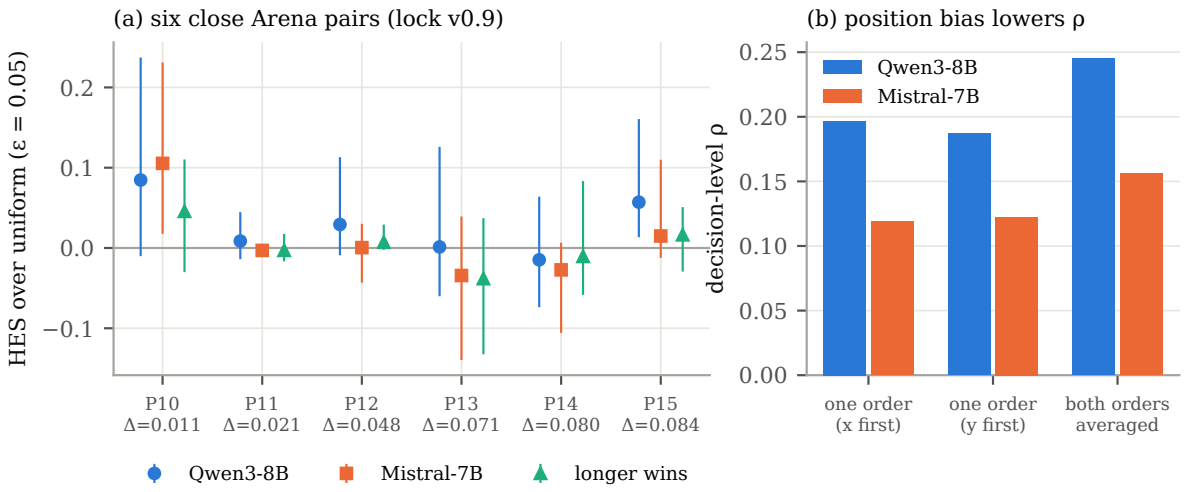


Figure 6: (a) HES over uniform sampling of three evaluators on six close LMArena pairs (Δ: human win-rate gap; pilot 50, $\varepsilon = 0.05$; 95% intervals). (b) Decision-level ρ with one presentation order and with both orders averaged.